

UNIVERSITI KEBANGSAAN MALAYSIA
*The National University of Malaysia*

# Lecture 1 Animations: Student Guide

**KKKC6353 Nanoelectronic Devices — learn by experimenting**

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October 2026

## How to use this guide

Eleven interactive animations go with the Lecture 1 notes. They are at **faizverse.github.io/nanoelectronic-devices/L1** and work on a laptop or phone, online or saved for offline use.

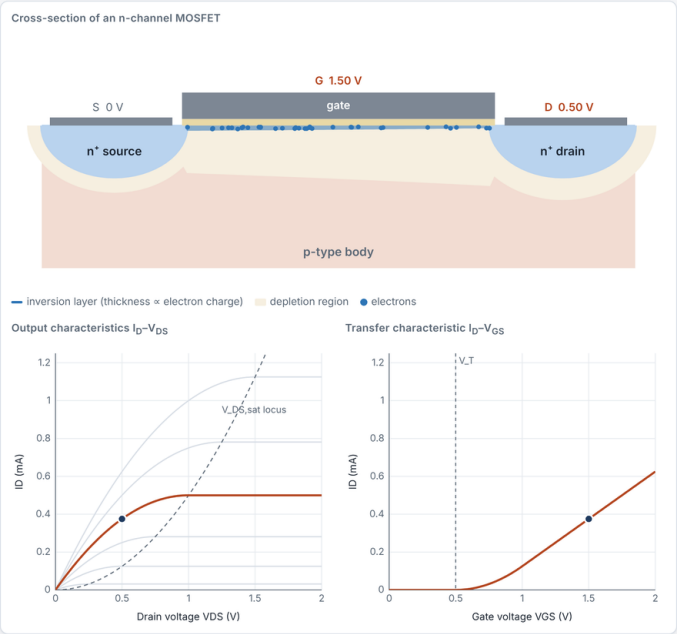
For each animation:

1. **Read first.** Skim the notes section listed, then read the *What you are seeing* box on the page.
2. **Predict, try, explain.** Before you move a slider, write down what you expect. Then try it. If the result surprised you, work out why.
3. **Answer the questions** at the end of each page in this guide, then check your answers under *Check yourself* on the web page.
4. **Note the big idea** in one sentence of your own.

Allow about 10 minutes per animation. Work through them in order: each builds on the one before.

| # | Animation | Read first | Link |
|---|---|---|---|
| 1 | MOSFET basics | Doc 2, § 3 | mosfet-basics |
| 2 | Natural length and threshold roll-off | Doc 3, § 1.1 and 2.1 | vt-rolloff |
| 3 | Drain-induced barrier lowering (DIBL) | Doc 3, § 2.2 | dibl |
| 4 | Punchthrough | Doc 3, § 2.3 | punchthrough |
| 5 | Subthreshold swing degradation | Doc 3, § 2.4 | subthreshold-swing |
| 6 | Channel-length modulation | Doc 3, § 2.5 | channel-length-modulation |
| 7 | Carrier velocity saturation | Doc 3, § 3.1 · Doc 2, § 2.4 | velocity-saturation |
| 8 | Hot carriers and impact ionisation | Doc 3, § 3.3 and 3.4 | hot-carriers |
| 9 | Oxide breakdown (TDDB) | Doc 3, § 3.4 | oxide-breakdown |
| 10 | Gate-induced drain leakage (GIDL) | Doc 3, § 4 | gidl |
| 11 | Variability: random dopants | Doc 3, § 5 | variability |

# 1 MOSFET basics




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/mosfet-basics.html
**Read first:** Document 2, Section 3

**What it shows.** Build the inversion channel with the gate, then push current through it with the drain. See why the current rises, then saturates, and where the channel pinches off.

> **The big idea.** The gate creates the channel; the drain drives current through it until the channel pinches off at $V_{DS} = V_{GS} - V_T$.

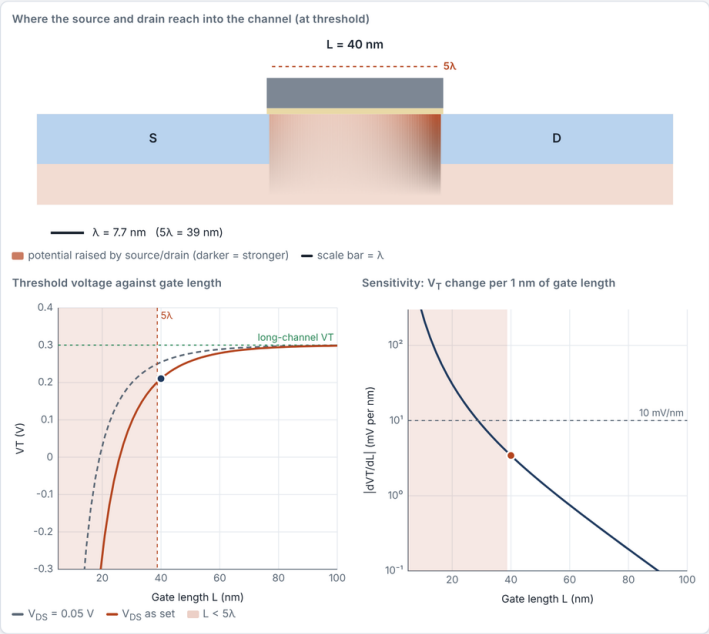
**Try this**

1. Set $V_{DS}$ = 0.1 V and raise $V_{GS}$ slowly from 0. At what $V_{GS}$ does current start to flow? What appears in the cross-section?
2. Set $V_{GS}$ = 1.5 V. **Predict** the $V_{DS}$ at which the channel pinches off, then raise $V_{DS}$ to check.
3. Keep raising $V_{DS}$ past pinch-off. What happens to the current? Where does the extra voltage go?
4. Compare the saturation current at $V_{GS}$ = 1.0 V and 1.5 V. What ratio do you get, and why?

**Questions** (check your answers on the page, under *Check yourself*)

- Why does the current saturate even though $V_{DS}$ keeps rising?
- Pinch-off means zero electron charge at the drain end. How can current still flow?
- Below $V_T$ this model gives $I_D = 0$. Is that true?

## 2 Natural length and threshold roll-off




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/vt-rolloff.html
**Read first:** Document 3, Sections 1.1 and 2.1

**What it shows.** How far does the drain's field reach into the channel? The natural length λ answers it, and sets how short a gate can be before the threshold voltage starts to fall away.

> **The big idea.** The natural length λ sets how short a gate can be: $L \geq 5\lambda$. Adding gates shrinks λ more than thinning anything does.

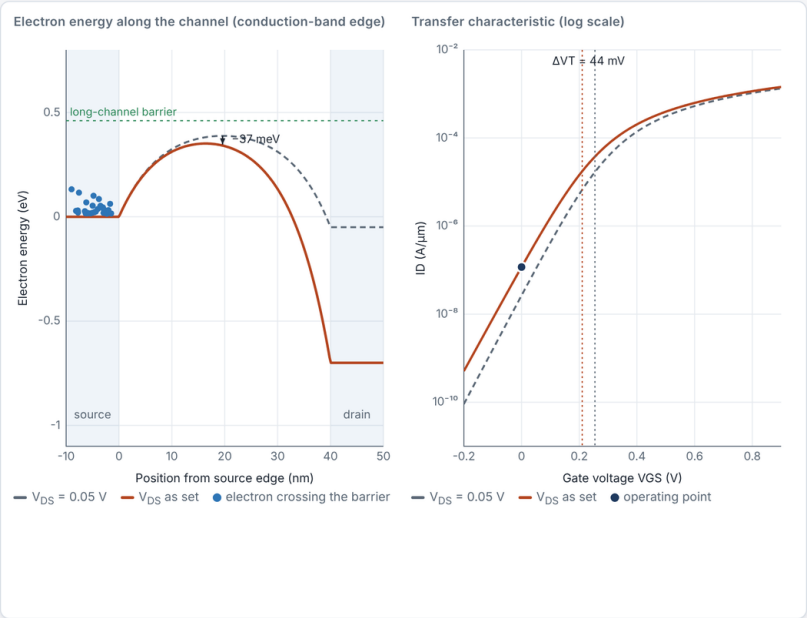
**Try this**

1. Planar bulk: move $L$ from 100 nm down to 20 nm. At roughly what $L$ do the source and drain zones meet and $V_T$ start to fall?
2. At $L$ = 30 nm, read the $V_T$ change per nm. If lithography varies $L$ by ±1 nm, how much does $V_T$ vary?
3. Halve the oxide thickness, then halve the body thickness. By how much does λ change each time?
4. Switch to FinFET, then GAA. What is the minimum gate length (≈ 5λ) for each?

**Questions** (check your answers on the page, under *Check yourself*)

- Why is changing the geometry (N) a stronger lever than thinning the oxide?
- Roll-off uses $V_{DS}$ = 0.05 V. Why does the high-$V_{DS}$ curve roll off earlier?
- Textbooks explain roll-off by "charge sharing". Is that wrong?

## 3 Drain-induced barrier lowering (DIBL)




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/dibl.html
**Read first:** Document 3, Section 2.2

**What it shows.** The drain reaches across a short channel and pulls down the barrier the gate is meant to control. Watch the barrier, the electrons that get over it, and the threshold shift that results.

> **The big idea.** In a short channel the drain pulls down the source barrier, so $V_T$ falls as $V_{DS}$ rises.

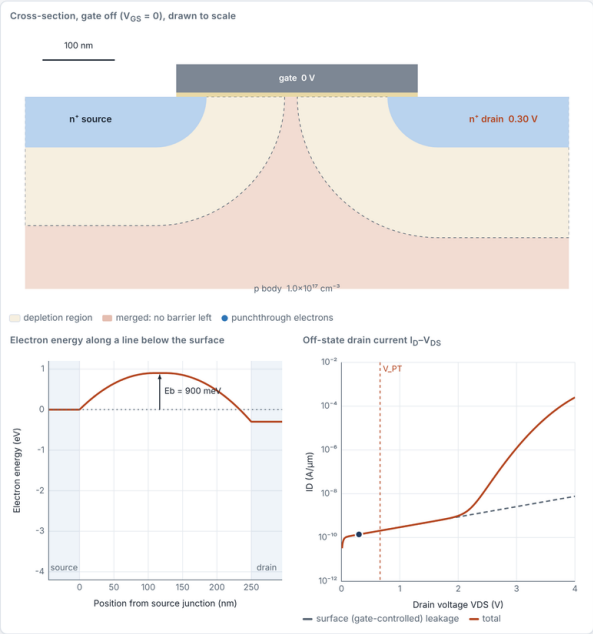
**Try this**

1. Planar bulk, $L$ = 100 nm, $V_{GS}$ = 0. Raise $V_{DS}$ from 0.05 to 1 V. Does the barrier move?
2. Repeat at $L \approx 40$ nm. Record the barrier drop and the DIBL value in mV/V.
3. Go to $L$ = 25 nm. Describe the barrier and read the off-current.
4. Keep $L$ = 25 nm and switch to FinFET, then GAA. **Predict** first: does DIBL rise or fall?

**Questions** (check your answers on the page, under *Check yourself*)

- Why does DIBL shift the transfer curve sideways rather than change its slope?
- Why does the barrier top move toward the source at high drain bias?
- A device has DIBL = 150 mV/V and SS = 90 mV/dec. How much larger is $I_{OFF}$ at $V_{DS}$ = 0.7 V than at 0.05 V?

# 4 Punchthrough




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/punchthrough.html
**Read first:** Document 3, Section 2.3

**What it shows.** Raise the drain voltage until the source and drain depletion regions meet below the surface. The barrier vanishes where the gate cannot reach, and current flows through the bulk.

> **The big idea.** When the source and drain depletion regions merge below the surface, current flows where the gate has no control.

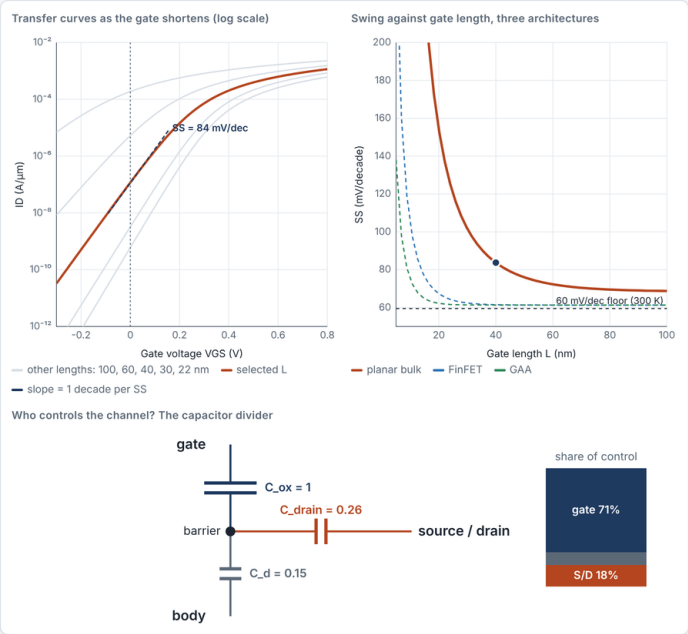
**Try this**

1. Use the defaults ($L$ = 250 nm, $10^{17}$ cm$^{-3}$). Raise $V_{DS}$ slowly until the depletion regions touch. Record $V_{PT}$.
2. Go a little past $V_{PT}$. How much does the current rise, and along which path does it flow?
3. Shorten $L$ to 120 nm. What is the new $V_{PT}$?
4. Raise the body doping to $5\times10^{17}$ cm$^{-3}$. What happens to $V_{PT}$? What might higher doping cost?

**Questions** (check your answers on the page, under *Check yourself*)

- Why can't the gate stop punchthrough current?
- Raising $N_A$ everywhere suppresses punchthrough. What does it cost?
- Is punchthrough a different effect from DIBL?

# 5 Subthreshold swing degradation




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/subthreshold-swing.html
**Read first:** Document 3, Section 2.4

**What it shows.** How many millivolts of gate voltage does it take to change the current tenfold? In a short channel the drain competes with the gate for control of the channel, and the answer gets worse.

> **The big idea.** The gate shares control of the barrier with the body and the drain. Even perfect control gives 60 mV/decade at 300 K.

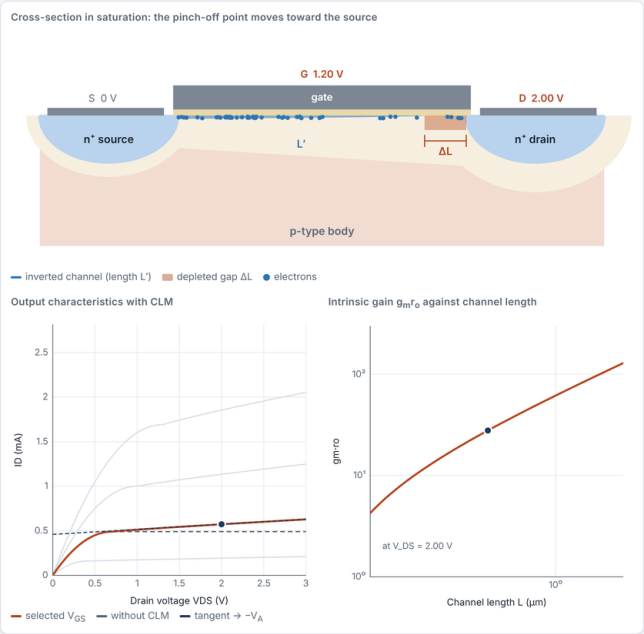
**Try this**

1. Planar bulk, $L$ = 100 nm. Read the swing (SS) and the gate's share of control.
2. Reduce $L$ to 25 nm. Watch the drain capacitor grow. Record SS and $I_{OFF}$.
3. Switch to FinFET, then GAA, at the same $L$. Which capacitor shrinks?
4. Try every setting to get below 60 mV/decade. Can you? Explain why not.

**Questions** (check your answers on the page, under *Check yourself*)

- Why can't any MOSFET beat 60 mV/dec at room temperature?
- A chip needs $I_{ON}/I_{OFF} = 10^5$ with $V_{DD}$ = 0.7 V. What is the largest SS it can tolerate if $V_T$ = 0.3 V?
- Why does the multi-gate body have almost no $C_d$?

## 6 Channel-length modulation




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/channel-length-modulation.html
**Read first:** Document 3, Section 2.5

**What it shows.** Past pinch-off, extra drain voltage widens the depleted gap at the drain and shortens the channel. The current keeps creeping up, and the transistor's analogue gain falls.

> **The big idea.** Beyond pinch-off the effective channel shortens, so current keeps rising. This is worse in short channels and lowers analogue gain.

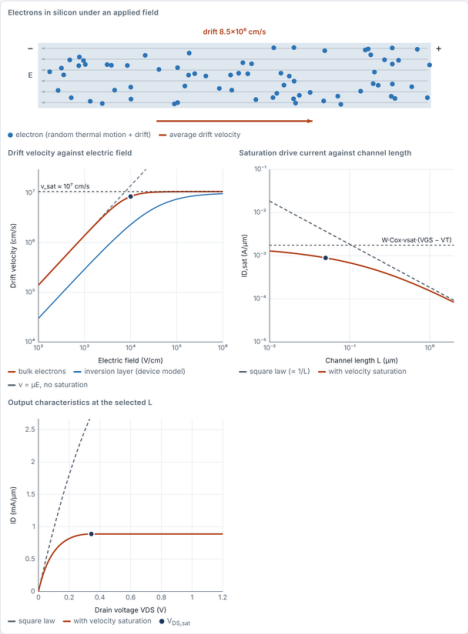
**Try this**

1. $L$ = 1 µm, $V_{GS}$ = 1.2 V. Raise $V_{DS}$ from 0 to 3 V and watch the pinch-off point move.
2. Read the gap ΔL and the current increase at 3 V. Repeat at $L$ = 0.25 µm.
3. Compare the Early voltage and the intrinsic gain at the two lengths.
4. Raise the body doping. **Predict** first: does ΔL grow or shrink?

**Questions** (check your answers on the page, under *Check yourself*)

- Why is CLM mainly an analogue concern?
- An analogue designer wants $g_m r_o \geq 50$ at $V_{ov}$ = 0.2 V. Roughly what $V_A$ does that need?
- Does the electron current really flow through a "depleted" region?

# 7 Carrier velocity saturation




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/velocity-saturation.html
**Read first:** Document 3, Section 3.1 · Document 2, Section 2.4

**What it shows.** Push harder on an electron and, beyond about $10^4$ V/cm, it stops going faster. In a short channel that caps the drive current and breaks the square law.

> **The big idea.** Electrons cannot drift much faster than about $10^7$ cm/s, so short-channel current is linear in overdrive and stops gaining from a shorter $L$.

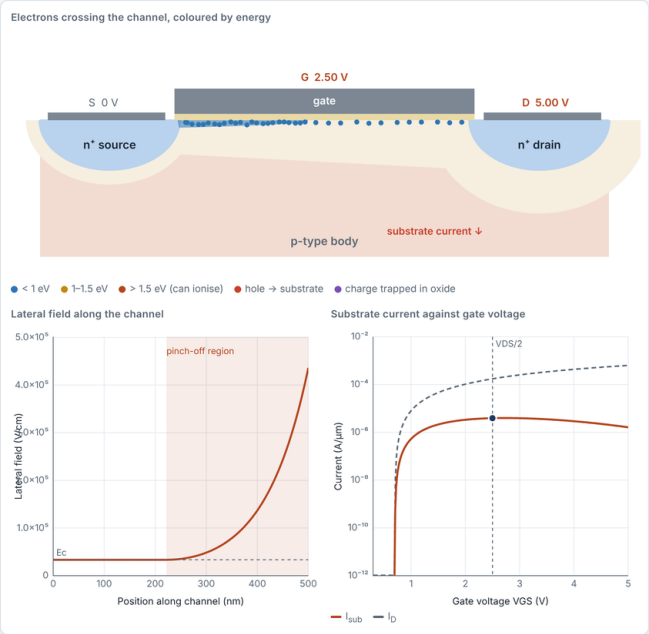
**Try this**

1. Set the field to $10^3$ V/cm. Compare the drift with the random thermal motion.
2. Raise the field to $10^4$, then $10^5$ V/cm. Where does the velocity stop increasing?
3. Move $L$ from 2 μm down to 20 nm. How does the saturation current change?
4. At $L$ = 30 nm, double the overdrive. Does the current double or quadruple?

**Questions** (check your answers on the page, under *Check yourself*)

- If current no longer improves with shorter L, why keep scaling L?
- Why does $V_{DS,sat}$ fall below $V_{GS} - V_T$ in a short channel?
- Holes saturate at a similar velocity but have lower mobility. What does that imply for p-MOS?

# 8 Hot carriers and impact ionisation




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/hot-carriers.html
**Read first:** Document 3, Sections 3.3 and 3.4

**What it shows.** Electrons pick up energy in the high-field region at the drain. A few gain enough to knock loose electron–hole pairs or jump into the gate oxide. See why this ruled the 5 V era and fades at 0.7 V.

> **The big idea.** Hot-carrier damage needs carriers with more than about 1.5–3.1 eV. From a 0.7 V supply they cannot gain that much energy.

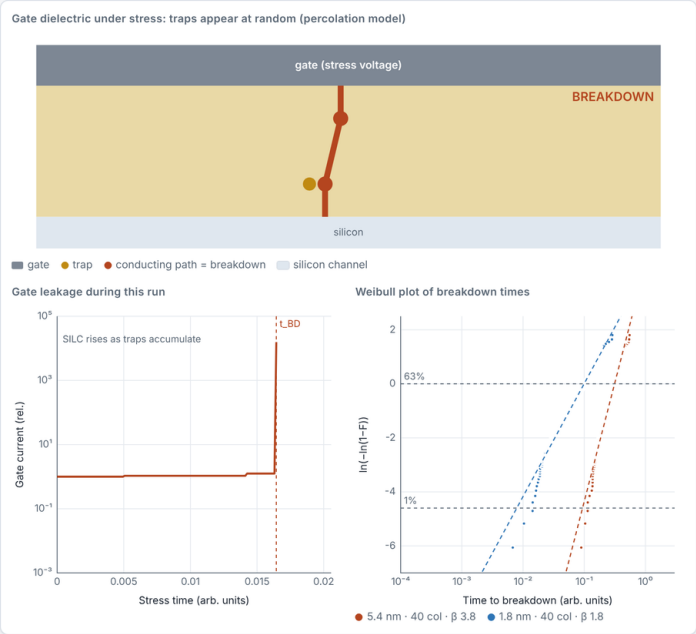
**Try this**

1. Choose 1990 · 5 V. Watch the electron colours near the drain. Look for holes and oxide injections.
2. Sweep $V_{GS}$ and find the peak substrate current. Compare its position with $V_{DS}/2$.
3. Turn on the lightly doped drain (LDD). How do the peak field and $I_{sub}$ change?
4. Choose Today · 0.7 V. The field is still high, so why does nothing get hot?

**Questions** (check your answers on the page, under *Check yourself*)

- The peak field in the 0.7 V device is similar to the 5 V device. Why no impact ionisation?
- Why does substrate current matter beyond reliability?
- How did the LDD help, and what did it cost?

# 9 Oxide breakdown (TDDB)




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/oxide-breakdown.html
**Read first:** Document 3, Section 3.4

**What it shows.** Under stress, defects appear at random in the gate dielectric. When a chain of them links gate to channel, the oxide breaks down. Run it once, then run it hundreds of times to see why breakdown is statistical and why thin oxides are fragile.

> **The big idea.** Oxide breakdown is random. Thin oxides fail earlier and with a wider spread.

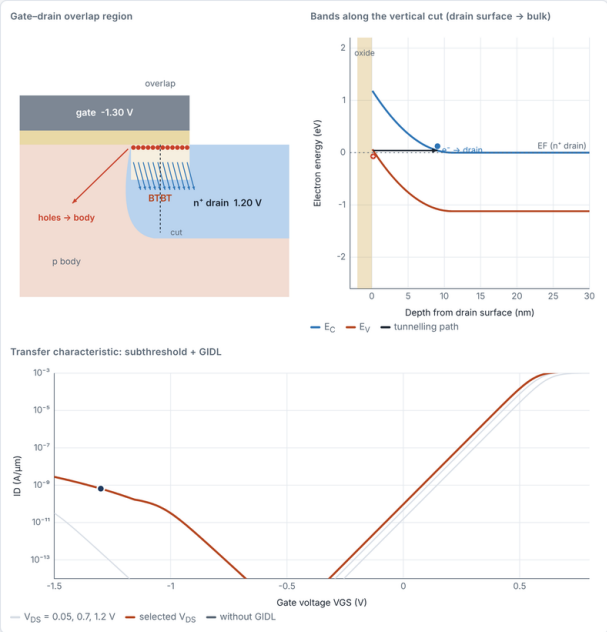
**Try this**

1. Thickness 5.4 nm. Stress one device and watch the traps build up until breakdown. Note $t_{BD}$.
2. Run it again. Is $t_{BD}$ the same? Why not?
3. Test 300 devices. Read the Weibull slope β.
4. Repeat at 1.8 nm. Compare where the line sits and how steep it is.

**Questions** (check your answers on the page, under *Check yourself*)

- Why does a larger gate area fail sooner?
- Why does a shallow Weibull slope (small β) worry a reliability engineer?
- Thin $SiO_2$ breakdowns are often “soft”. What does that mean?

# 10 Gate-induced drain leakage (GIDL)




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/gidl.html
**Read first:** Document 3, Section 4

**What it shows.** Turn the gate hard off with the drain high, and the field across the gate–drain overlap bends the bands so steeply that electrons tunnel straight from the valence band into the conduction band. Off-current then rises as the gate goes more negative.

> **The big idea.** A strong gate–drain field lets electrons tunnel band to band. This sets a floor under the off-current.

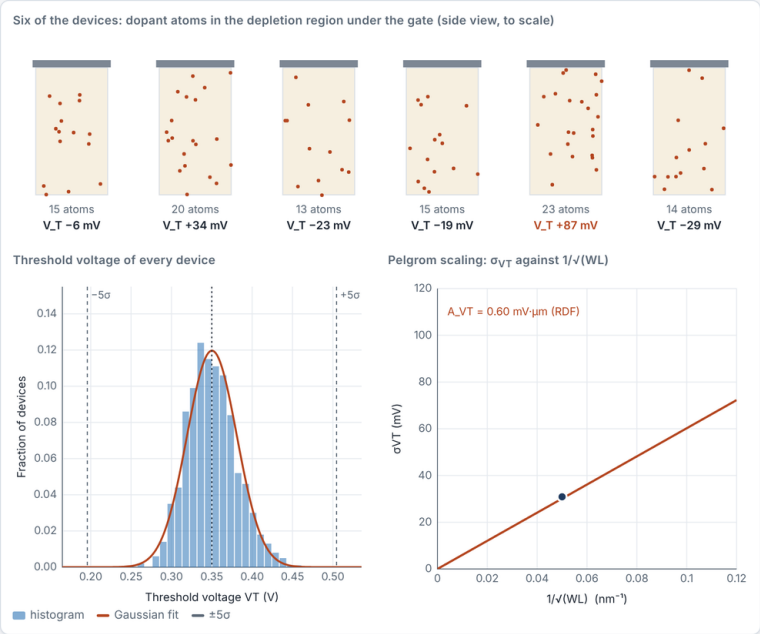
**Try this**

1. $V_{DS}$ = 1.2 V, EOT 2 nm. Move $V_{GS}$ from +0.3 V to −1.5 V. When do the bands bend enough for tunnelling?
2. Find the minimum of the transfer curve. Why does the current rise again at negative $V_{GS}$?
3. Thin the oxide to 1 nm. Where does GIDL start now?
4. Lower $V_{DS}$ to 0.7 V. How much does GIDL drop?

**Questions** (check your answers on the page, under *Check yourself*)

- Why does a thinner gate dielectric make GIDL worse?
- Why does the band bending stop growing near 1.2 V?
- Is GIDL a short-channel effect?

## 11 Variability: random dopants




**Link:** https://faizverse.github.io/nanoelectronic-devices/L1/variability.html
**Read first:** Document 3, Section 5

**What it shows.** Fabricate a thousand nominally identical transistors. In a small channel there are only a handful of dopant atoms, their number and positions are random, and so is every device's threshold voltage.

> **The big idea.** Small channels hold only a few dopant atoms, so $V_T$ varies randomly as $1/\sqrt{(WL)}$. Undoped channels remove most of this spread.

**Try this**

1. $W = L = 200$ nm. Fabricate 1000 devices. Note the dopants per device and $\sigma_{VT}$.
2. Shrink to 20 × 20 nm and fabricate again. Count the atoms in the sample devices and record $\sigma_{VT}$.
3. Halve the oxide thickness. What happens to $\sigma_{VT}$?
4. Choose the undoped channel. What variability is left?

**Questions** (check your answers on the page, under *Check yourself*)

- Why does an atom near the surface affect $V_T$ more than a deep one?
- Doubling both W and L — what happens to $\sigma_{VT}$?
- If RDF disappears in undoped channels, why is variability still a problem?